

Python Lab Examination

Product Inventory Management System

Duration: 2 Hours

Maximum Marks: 40

Objective

In this lab examination, you are required to develop a basic Python application using fundamental programming concepts.

This task evaluates your ability to:

- Use variables, conditions, and loops
 - Work with **lists and dictionaries**
 - Implement simple functions
 - Perform basic CRUD operations
 - Handle simple input validation
-

Problem Statement

Develop a “**Product Inventory Management System**” using Python.

The application should allow users to:

- Add products
 - View all products
 - Search products
 - Update products
 - Delete products
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Data Storage (Important Change)

- Data must be stored in a **list of dictionaries**
- Each product is represented as a dictionary

Example Structure

```
products = [  
    {"id": 1, "name": "Laptop", "category": "Electronics", "price": 55000, "quantity": 10},  
    {"id": 2, "name": "Chair", "category": "Furniture", "price": 1500, "quantity": 50}  
]
```

Product Fields

Each product must contain:

- **id** (auto-generated integer)
- **name** (string)
- **category** (string)
- **price** (float)
- **quantity** (int)

Important:

- **id** should be **auto-generated** (incremental)
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Sample Data (Optional for Testing)

ID	Name	Category	Price	Quantity
1	Laptop	Electronics	55000	10
2	Smartphone	Electronics	20000	25
3	Chair	Furniture	1500	50
4	Notebook	Stationery	50	200
5	Bottle	Accessories	300	80

Menu-Driven Program

Your program must display the following menu repeatedly:

1. Add Product
2. View All Products
3. Search Product
4. Update Product
5. Delete Product
6. Exit

- The program should run until the user selects **Exit**
 - Handle invalid menu choices properly
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Functional Requirements

1. Add Product

- Accept:
 - name
 - category
 - price
 - quantity
 - Auto-generate **id**
 - Add product to list
-

2. View All Products

- Display all products in a clear format
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3. Search Product

Search using:

- Product ID
- OR Name (simple match)

Display matching records or a message if not found

4. Update Product

- Ask for product ID
 - Allow updating:
 - name
 - category
 - price
 - quantity
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5. Delete Product

- Remove product using product ID
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Basic Validation Rules

Apply simple validation:

- Name → cannot be empty
- Category → cannot be empty
- Price → must be > 0
- Quantity → must be ≥ 0

If invalid:

- Show error message
 - Ask user to re-enter
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Exception Handling

- Handle invalid inputs using `try-except`
 - Program should not crash
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Evaluation Scheme (40 Marks)

Component	Marks
Menu-driven program	8
Add/View operations	8
Update/Delete operations	8
Search functionality	6
Use of list & dictionary	4
Input validation	3
Exception handling	3

Expected Outcome

A correct program should:

- Store data in memory using list of dictionaries
- Perform all CRUD operations correctly
- Handle invalid input without crashing
- Display clean and readable output